

Recalling (Some) U.S. Vice Presidents

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Abstract

Undergraduates (n = 118) recalled as many U.S. Vice Presidents (VPs) as they could in 5 minutes. Only exact position names were counted as correct. The typical serial position effect (SPE) curve emerged with more recent VPs being recalled than earlier VPs. This pattern also emerged when the 49 VPs were grouped 3 at a time. These patterns partially replicate Healy and Parker (2001). Recall of VP # 48 (Pence) and VP #49 (Harris) showed decreased recall despite being the 2 most recent VPs. Several VPs who went on to be the President of the United States (Adams, Jefferson, T. Roosevelt, Nixon, L. Johnson, Bush and Biden) enjoyed high overall proportion recall (.3845). Results are discussed.

Key Words: Serial Position Effect, Vice-President, Collective Memory, Information Processing, Cognition.

Recalling (some) U.S. Vice Presidents

For more than a century (Ebbinghaus, 1913), the serial position effect (SPE) has been a standard and replicable memory phenomenon. SPE performance (usually free recall) is observed when the first few and last few items in a series (numbers, letters, words, etc.) are remembered best, while items in the middle of the series are remembered worst. The SPE contains two additional effects: The primacy effect (first few series items are remembered better than middle series items) and the recency effect (last few items are remembered better than the middle and first few series items). One type of memory (collective memory) frequently involves using the SPE paradigm. Collective memory refers to the way that a group of people remember past events. Previous research has shown the presence of collective memory in Americans' knowledge of U.S. presidents (DeSoto & Roediger, 2019). Data showed a large primacy and recency effect, as well as increases in remembrance of particularly influential presidents (e.g. Abraham Lincoln). These results are also not confined to the U.S. leaders, as evidenced by the work of Fu, Xue and colleagues (2016). While many such findings have been around for decades (Brown & Siegler, 1991; Roediger & Crowder, 1976), very little has examined recall of U.S. Vice Presidents (VPs). A review could only find one publication (Healy & Parker, 2001) that examined SPE's of the 44 VPs. These authors also examined recall of the presidents as well. They found a) the typical bow-shaped SPE curve for recall of both presidents and VP and b) overall recall performance was greater for presidents than VPs. They surmised that factors including frequency of exposure and distinctiveness (see Neath, 1993a, b) may likely be relevant to the effects they observed. One reason that paper may be the only one to examine VP recall may be that, for the most part, results of U.S. president recall typically reveal similar, replicable findings. Large primacy (G. Washington) and recency (J. Biden) effects, in addition to the recall bump often found for the 16th U.S. president (Lincoln) are repeated across studies. In other words, president recall studies are highly replicable and, presidents are often seen as more important, more relevant and even more popular than VPs. This view may be changing, however, with the current (49th VP), Kamala Harris. Harris may be perceived as distinctive because she is the first female, African-American/South Asian VP in American History. Harris is currently running for President. If these factors are seen as distinctive, Harris' proportion recall should reflect this.

In the present study the goal was to attempt a partial replication of Healy & Parker (2001). In 2001, there were only 44 VPs (Gore, Cheney, Biden, Pence, & Harris have since been added). Arguably, and in terms of Crowder's (1993) position that distinctiveness is a prime (but not the only) factor in SPE recall, these 5 VPs may display some level of distinctiveness. Harris is also running for President but wasn't when the current study was performed. Also, Biden's recall could be due to him being a VP as well as currently being President. Likewise, Pence's work during COVID may increase his recall performance. Furthermore, because our participants are young, they may be less likely to recall more remote VPs. In this case distinctiveness and recency effects may be more informed by the fact that these participants likely directly witnessed campaigns, they may have voted in recent elections and are living under more recent administrations.

H1 – It is expected that this experiment will partially replicate Healy and Parker (2001) in that there will be robust primacy and recency effects, when examining VP recall as a function of position on the list (1-49). The recency effects may be more pronounced (higher recall proportion) due to the distinctiveness of Harris’ accomplishments.

H2 - It is also expected that a lower overall recall performance for VPs will occur, also in line with Healy and Parker (2001). Finally, like Healy and Parker (2001), a plot of triplets (combining 3 VPs into 1 recall proportion) should reveal a similar SPE curve. Because 49 is not divisible three, the last triplet will actually be the average proportion recall of VPs 46-49.

Method

Sample

A total of 118 undergraduates (53 male, 65 female, all US citizens) participated in a classroom demonstration of the SPE. A total of 111 identified as Gen Z (born between 1997-2012) with the remaining 7 identifying as Millennials (born between 1981-1996). Exact age was not recorded. Education level ranged from 12 years (freshman) to 16 years (senior) and averaged 15.45 years. Although exact age was not measured, it is estimated that the average age of all participants was approximately 23.5 years.

Materials & Procedures

The University’s Institutional Review Board approval was obtained before the study started. Testing was done in a group setting. A Qualtrics link was posted on a projector screen at the front of the room. Accessing the link opened a consent form and instructions. A statement appeared asking if they wanted to continue. If they agreed this constituted informed consent. Participants were then asked their gender, education level and whether they were Baby Boomers, Gen-X, or Millennials. They then were instructed that 49 blanks would appear, and they were to type in the name of the U.S. vice president that corresponded to each blank. If they could not recall a VP’s name, they were instructed to move on to the next blank. They could not return to any blanks they did not fill in. When finished they were thanked and debriefed of the study’s purpose. All completed the study within 5 minutes. Data were collected in May of 2021.

Results

The scoring protocol was strict and only the correct VP for lines 1 through 49 was counted as correct. Since three VPs had the last name Johnson (R. Johnson, A. Johnson, L. Johnson) these only counted as correct if the first initial was present. Table 1 lists VP name as a function of VP position (1-49) and includes total number of participants out of 118 who had the correct VP listed in the correct position, based on the scoring protocol detailed previously. Also listed are the triplet averages (in sets of 3 scores for positions 1-45 and 4 scores for positions 46-49). Figures 1 and 2 display these results.

First, both Figure 1 (proportion recall as a function of VP position) and Figure 2 (proportion recall using the triplet position procedure) of Healy and Parker (2001, p. 177, Figure 9.1) showed a similar SPE with greater proportion recall for recent VPs versus earlier VPs. Second, although we didn’t obtain SPEs for U.S. Presidents in the present study, our overall proportion correct response performance for the 49 VPs was lower than overall proportion correct response performance for presidents. These two results offer partial replication of Healy and Parker (2001), and only for their proportion of correct response data. Unlike that prior study, Healy and Parker (2001), the current one did not gather familiarity ratings or categorization responses. Third, and contrary to what was predicted, the 47th VP (Biden) proportion recall (.763) was higher than the proportion recall of the 48th VP (Pence, .746) and the 49th VP (Harris, .389). Fourth, Table 1 lists those VPs who obtained proportion recall correct responses greater than 25% (chosen somewhat arbitrarily). This list includes the 1st VP (Adams), 2nd (Jefferson), 25th (Roosevelt), 36th Nixon), 43rd (Bush), and 45th, 47th, 48th, and 49th VPs (Gore, Biden, Pence, and Harris). All remaining VPs had proportion recalls ranging from .042 (King, Breckinridge, Stevenson, Dawes, Garner, Wallace, & Barkley) to .229 (Truman). Table 1 also lists those VPs who eventually became a U.S. President (Adams,

Table 1*Number Correct, Proportion Correct and Mean Triplet Performance as a Function of the 49 Vice Presidents*

	VP	# Correct	# Correct/118 (proportion)	M Triplet # correct	M Triplet # correct/118 (proportion)
1	Adams	70	.593*#		
2	Jefferson	53	.449*#		
3	Burr	26	.220	49.67	.421
4	Clinton	16	.136		
5	Gerry	10	.085		
6	Tompkins	7	.059	11.00	.093
7	Calhoun	13	.110		
8	Van Buren	14	.119		
9	R. Johnson	8	.068	11.67	.099
10	Tyler	14	.119		
11	Dallas	7	.059		
12	Fillmore	12	.102	11.00	.093
13	King	5	.042		
14	Breckinridge	5	.042		
15	Hamlin	6	.051	5.33	.045
16	A. Johnson	19	.161		
17	Colfax	5	.042		
18	Wilson	9	.076	11.00	.093
19	Wheeler	7	.059		
20	Arthur	13	.110		
21	Hendricks	6	.051	8.67	.073
22	Morton	7	.059		
23	Stevenson	5	.042		
24	Hobart	6	.051	6.00	.051
25	T. Roosevelt	30	.254*#		
26	Fairbanks	8	.068		
27	Sherman	6	.051	14.67	.124
28	Marshall	6	.051		
29	Coolidge	14	.119		
30	Dawes	5	.042	8.33	.071
31	Curtis	7	.059		
32	Garner	5	.042		
33	Wallace	5	.042	5.67	.048
34	Truman	27	.229#		
35	Barkley	5	.042		
36	Nixon	30	.254*#	20.67	.175
37	L. Johnson	26	.220#		
38	Humphrey	8	.068		
39	Agnew	8	.068	14.00	.119
40	Ford	22	.186		
41	Rockefeller	9	.076		
42	Mondale	7	.059	12.67	.107
43	Bush	37	.314*#		
44	Quale	9	.076		
45	Gore	32	.271*	26.00	.220

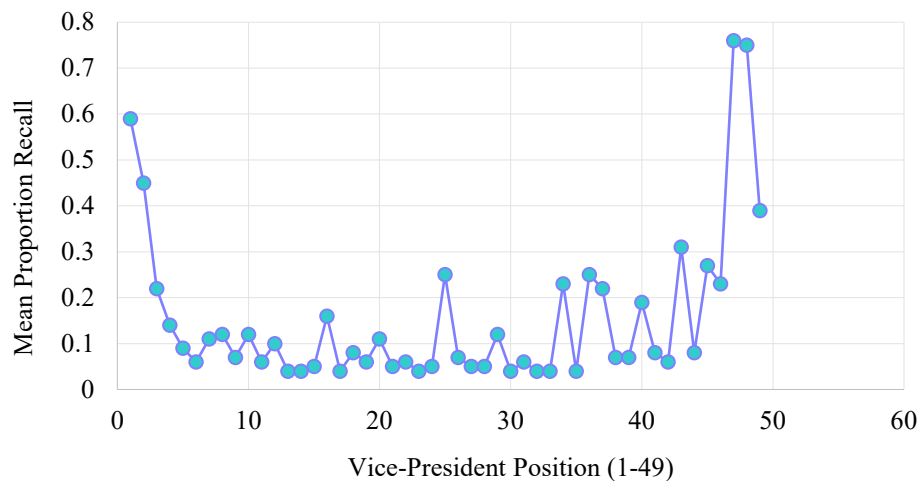
Table 1 (Continued)

46	Cheney	27	.229		
47	Biden	90	.763*#		
48	Pence	88	.746*		
49	Harris	46	.389*	62.75	.531

Notes: * = proportion recall > .250; # = future U.S. President

Figure 1

Mean Proportion Recall as a Function of Vice-President Position (1–49)



Jefferson, Roosevelt, Truman, Nixon, L. Johnson, Bush, & Biden). Also noteworthy is that the proportion recall for VP # 25 (T. Roosevelt), is similar to the often observed recall for the 16th President (Lincoln).

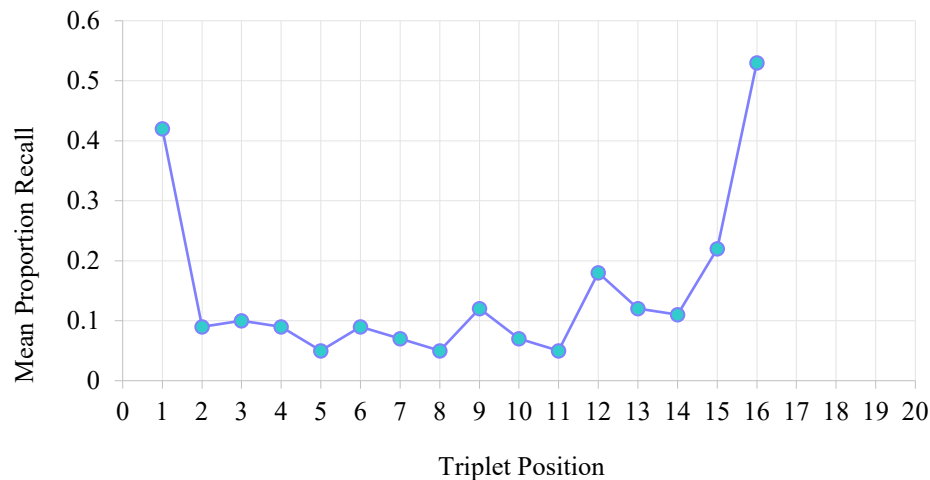
Discussion

A partial replication of Healy and Parker (2001) was obtained with regard to recall of U.S. VPs. Their overall VP recall was lower than their overall President of the U.S. recall. While we didn't gather U.S. President recall, our VP recall is lower than their President recall. The present results also conformed to the typical SPE curve with lower recall for earlier (primacy effect) VPs than current VPs (recency effect). Recall for VP # 48 (Pence) and VP # 49 (Harris) was lower than recall for VP # 47 (Biden). This was not expected for a number of reasons. First, the vice-presidency of Pence was arguably the most contentious in recent memory. Likewise, with Harris' many firsts, one would have expected higher recall. This is not to say the recall of Pence and Harris was not exceptional (.746 for Pence, .389 for Harris). The present data were collected in May of 2021, so their vice-presidencies were (presumably) quite recent in the minds of the participants at the time of this experiment. Second, Harris' firsts were reported daily and would presumably have become reinforced over time. That said, it may also be the case that our participants (all college students) simply were not paying as much attention to Harris' vice-presidency. Note that the pandemic was on-going during late 2019 and early 2021, which may have distracted participants, with them paying less attention to this event. There is evidence (Glanzer & Cunitz, 1966) that adding a delay between presentation and recall of items in short-term memory reduces the typical recency effect.

Another possible account involves distinctiveness (Neath, 1993a, b), which Neath suggests is able to predict and explain many SPE effects both in recognition and recall. Increases or decreases in the amount of distinctiveness a stimulus possesses could impact the SPE curve. While it was expected in the present experiment that the typical SPE recency curve would have shown VP # 48 (Pence) and VP # 49 (Harris) to have a higher overall proportion correct score than Biden (VP # 47), that was not the case. Distinctiveness data were not obtained in the present experiment, nor familiarity ratings. This was due to the fact that the Healy and Parker (2001) article was found after the study was performed. It should be noted, however, that Pence's work during COVID and Harris' current run for President may lead to higher recall if the study were done in 2024, rather than in 2021.

Figure 2

Mean Proportion Recall as a Function of Triplet Position.



Notes: triplet position 16 was actually the mean of VPs 46-49.

One final observation is that some VPs were remembered better than other VPs (see Table 1, T. Roosevelt). The remaining 30 VPs' proportion recall for positions 3-44 ranged from .042 to .161. This pattern is similar to SPE proportion recall performance on U.S. Presidents, notably the 16th President (Lincoln; see DeSoto & Roediger, 2019, p. 140). But unlike Lincoln's effect on increasing recall of other U.S. President's positions before and after him, many VPs before or after Roosevelt were recalled poorly.

Limitations

The current experiment has several limitations. First, college students served as participants, thereby limiting generalizability. Second, and unlike the only other publication that has examined VP recall in a SPE paradigm (Healy & Parker, 2001), only correct proportion recall performance was examined. Third, the study did not ask the same participants to recall the current U.S. Presidents, like Healy and Parker (2001). Fourth, the duration of time (5 minutes) participants engaged in the experiment is quite brief, suggesting that little effort was exhibited in recalling VPs. Fifth, and as mentioned previously, our younger participants may be less likely to recall more remote VPs.

Conclusion

The current experiment partially replicated Healy and Parker (2001) and now becomes the 2nd to examine U.S. VP recall. These results add to the growing list of collective memory studies that have relied on the SPE paradigm. The relevance of our results within the collective memory field allows for the continued increase in group identity, political discourse and social information processing. Thus, such studies inform readers about societal issues (i.e., 9/11, WW II, dropping of the atomic bomb) as well as psychological issues (what do people remember and why do they remember it?). Although U.S. VP recall might not be ranked with 9/11 as the most relevant collective memory, U.S. VP recall does inform individuals and cultures about the impact of social events that guide and shape the future.

Statements and Declarations - The author reports no conflict of interest. Ethical approval was obtained by the respective institutional review board prior to data collection.

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